

Weekly Python Assignment 2 - Group 3 - S23

Instructions: For each exercise below, type the requested code into the cell below the exercise, then run your code to verify that it works correctly.

Color Code (nonfunctional in Google Colab):

- External File/Imported Module
- User-created Function
- Python/Imported Function
- Name-specific Data Structure/Variable

Relative Difficulty & Complexity

- **Easy (E):** Covered during the week's lecture. Typically single-step exercise
- **Medium (M):** Some combination with topics previously covered. Mostly double-step exercise
- **Hard (H):** Combination with topics previously covered. Multi-step exercise

Breakdown: 8 (E) & 2 (M)

Example 1: Assign 9 to the variable y. Then assign 10 to the variable x and print x, but only if y is equal to 9.

```
In [1]: y = 9
if y==9:
    x = 10
    print(x)
```

10

1(E). Have the user input an integer. If it is equal to 7 or 11, print "Jackpot!". If it is equal to 2 or 12, print "You Lose!". Otherwise, print "Roll again". Run your code and display the correct output for each possible case.

```
In [14]: #It's arranged nicely. If this is a py file, I won't have to do this.
#num = int(input("Enter an integer:")) will be on py file and does not need to write p
num = int(input(print("Enter an integer:", end = "")))

print(num)

if num == 7 or num == 11:
    print("Jackpot!")
elif num == 2 or num == 12:
    print("You Lose!")
else:
    print("Roll again")
```

Enter an integer:1
Roll again

```
In [12]: num = int(input(print("Enter an integer:", end = "")))

print(num)

if num == 7 or num == 11:
    print("Jackpot!")
elif num == 2 or num == 12:
    print("You Lose!")
else:
    print("Roll again")
```

Enter an integer:11
Jackpot!

```
In [13]: num = int(input(print("Enter an integer:", end = "")))

print(num)

if num == 7 or num == 11:
    print("Jackpot!")
elif num == 2 or num == 12:
    print("You Lose!")
else:
    print("Roll again")
```

Enter an integer:2
You Lose!

2(E). Ask the user to input the number of bags of cement they wish to purchase. If the number of bags is less than 10, the cost for each bag is \\$6.95 then print the total cost. If the number of bags is greater than or equal to 10, the cost for each bag is \\$5.95 then print the total cost. Remember to include units and to round to relevant precision.

```
In [17]: num_bags = int(input(print("Enter the number of bags:", end = "")))

print(num_bags)

if num_bags < 10:
    cost = 6.95
else:
    cost = 5.95

print("Total_price = $%.2f" %(num_bags * cost))
```

Enter the number of bags:5
Total_price = \$34.75

3(E). Set the variable `fastest_mode` to True. If `fastest_mode` is True, set `game_speed` to 11. Otherwise, set `game_speed` to 2. Then print `fastest_mode` and `game_speed`.

```
In [4]: fastest_mode = True

if fastest_mode == True:
    game_speed = 11
else:
    game_speed = 2
```

```
print("fastest_mode = %s, game_speed = %d" %(fastest_mode, game_speed))
```

```
fastest_mode = True, game_speed = 11
```

4(E). Import the `random` module, but abbreviate it as `r`. Print the functions contained in this module.

```
In [8]: import random as r
print(dir(r))
```

```
['BPF', 'LOG4', 'NV_MAGICCONST', 'RECIP_BPF', 'Random', 'SG_MAGICCONST', 'SystemRandom', 'TWOPI', '_ONE', '_Sequence', '_Set', '__all__', '__builtins__', '__cached__', '__doc__', '__file__', '__loader__', '__name__', '__package__', '__spec__', '_accumulate', '_acos', '_bisect', '_ceil', '_cos', '_e', '_exp', '_floor', '_index', '_inst', '_isfinite', '_log', '_os', '_pi', '_random', '_repeat', '_sha512', '_sin', '_sqrt', '_test', '_test_generator', '_urandom', '_warn', 'betavariate', 'choice', 'choices', 'expovariate', 'gammavariate', 'gauss', 'getrandbits', 'getstate', 'lognormvariate', 'normalvariate', 'paretovariate', 'randbytes', 'randint', 'random', 'randrange', 'sample', 'seed', 'setstate', 'shuffle', 'triangular', 'uniform', 'vonmisesvariate', 'weibullvariate']
```

5(E). Import just the square root function from the module `numpy`, abbreviated as `np`. Print the square root of 31.

```
In [10]: from numpy import sqrt as np
print(np(31))
```

```
5.5677643628300215
```

6(M). Use a While loop to print the odd numbers from 1 to 31, but skip any number that is evenly divisible by 3.

```
In [18]: a = 0
while a < 32:
    a += 1
    if a % 3 == 0 or a % 2 == 0:
        pass
    else:
        print(a)
```

```
1
5
7
11
13
17
19
23
25
29
31
```

7(E). Use a For loop with `range` to print all the multiples of 7 less than 100.

```
In [21]: for i in range (7, 100, 7):
          print(i)
```

```
7
14
21
28
35
42
49
56
63
70
77
84
91
98
```

8(E). Use a While loop to repeatedly ask a user to input a number of bags of cement. Each time the user submits a value, calculate and print the required number of bags of sand and gravel assuming a 1:2:3 ratio (a bag of cement is 3.0 cubic foot, but bags of sand and gravel are only 0.5 cubic foot. Also print the number of cubic feet of concrete this will make, using the information in the Engineering Assignment 1. End the loop when the user enters 0. Remember to include units and to round to relevant precision. Test your loop with 3 different values.

```
In [31]: while True:
          cement_bags = int(input(print("Enter a number of bags of cement:", end = "")))
          print(cement_bags)
          if cement_bags == 0:
              break
          sand_bags = 6 * 2 * cement_bags
          gravel_bags = 6 * 3 * cement_bags
          concrete_volume = cement_bags * 4.5 * 3 # 1cubic foot cement + 2cubic foot sand +

          print("cement bags = %d" %cement_bags)
          print("sand bags = %d" %sand_bags)
          print("gravel bags = %d" %gravel_bags)
          print("Total concrete = %.2fft3" %concrete_volume)
```

```
Enter a number of bags of cement:4
cement bags = 4
sand bags = 48
gravel bags = 72
Total concrete = 54.00ft3
Enter a number of bags of cement:7
cement bags = 7
sand bags = 84
gravel bags = 126
Total concrete = 94.50ft3
Enter a number of bags of cement:10
cement bags = 10
sand bags = 120
gravel bags = 180
Total concrete = 135.00ft3
Enter a number of bags of cement:0
```

9(E). Use a For loop to count the number of vowels in the word "harmonica". Do not manually count.

```
In [35]: cnt = 0
vowels = ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U']
for i in "harmonica":
    if i in vowels:
        cnt += 1
print(cnt)
```

4

10(M). The Fibonacci numbers follow the sequence 1, 1, 2, 3, 5 where each number is the sum of the two previous numbers in the sequence. Use a For loop to calculate and print out the first 10 numbers in the Fibonacci sequence.

```
In [46]: box0 = 1
box1 = 0
box2 = 0
for i in range (10):
    if i == 0:
        print(box0)
        box1 = box0
    else:
        print(box1)
        box2 = box1 + box0
        box0 = box1
        box1 = box2
```

1
1
2
3
5
8
13
21
34
55